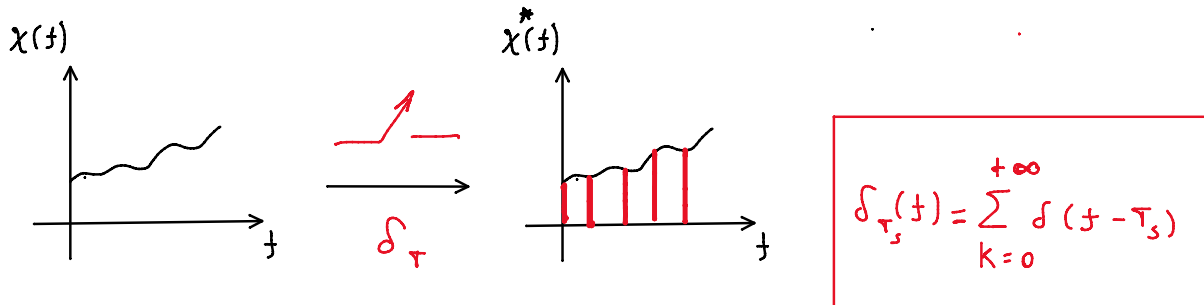


Impulse Sampling:



$$X(s) \xrightarrow{\quad} X^*(s)$$

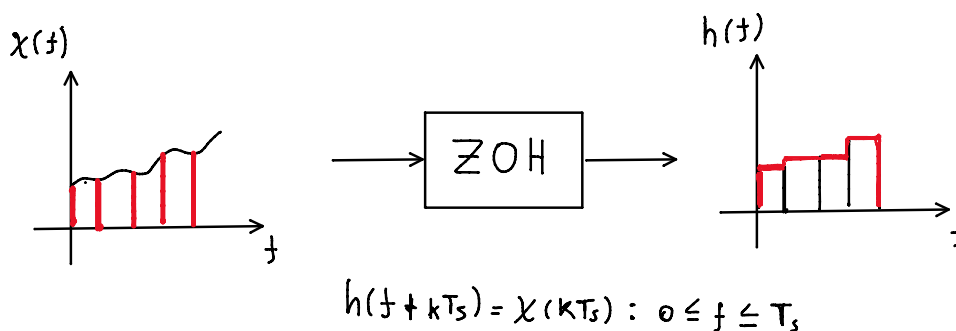
$$\begin{aligned} x^*(t) &= \sum_{k=0}^{+\infty} x(kT_s) \cdot \delta(t - kT_s) \\ &= x(0) \delta(t) + x(T_s) \delta(t - T_s) + \dots \end{aligned}$$

$$\begin{aligned} x^*(t) &= \sum_{k=0}^{+\infty} x(kT_s) \delta(t - kT_s) \xrightarrow{\mathcal{L}\{\}} X^*(s) = x(0) \mathcal{L}\{\delta(t)\} + x(T_s) \mathcal{L}\{\delta(t - T_s)\} + \dots \\ X^*(s) &= x(0) + x(T_s) \cdot e^{-T_s s} + x(2T_s) \cdot e^{-2T_s s} + \dots \end{aligned}$$

$$\begin{aligned} z^{-1} &= e^{-T_s s} \longrightarrow X^*(s) = \sum_{k=0}^{+\infty} x(kT_s) \cdot z^{-k} \\ s &= \frac{1}{T_s} \ln(z) \end{aligned}$$

$$\begin{aligned} X^*(s) \Big|_{s = \frac{1}{T_s} \ln(z)} &= X(z) \\ s &= \frac{1}{T_s} \ln(z) \end{aligned}$$

Data Hold:



$$h(t) = x(0) [u(t) - u(t - T_s)] + x(T_s) [u(t - T_s) - u(t - 2T_s)] + \dots$$

$$h(t) = \sum_{k=0}^{+\infty} x(kT_s) [u(t - kT_s) - u(t - (k+1)T_s)] \xrightarrow{\mathcal{L}\{\}} H(s) = \sum_{k=0}^{+\infty} x(kT_s) \cdot \left[\frac{e^{-kT_s s} - e^{-(k+1)T_s s}}{s} \right]$$

$$H(s) = \frac{1 - e^{-T_s s}}{s} \cdot \underbrace{\sum_{k=0}^{+\infty} x(kT_s) e^{-kT_s s}}_{X^*(s)} = G_{h_p}(s) \cdot X^*(s)$$

$$G_{h_p}(s) = \frac{1 - e^{-T_s s}}{s}$$

